

Math 289, Fall 2026

Lecture 2

Warm-up. If

$$x \wedge x \wedge x \wedge x \wedge x \cdots = 2,$$

where $x \wedge y = x^y$, what is x ?

Problem 1. Prove that $\sqrt{2} + \sqrt{3} + \sqrt{5}$ is an irrational number.

Problem 2. Show that no set of nine consecutive integers can be partitioned into two sets with the product of the elements of the first set equal to the product of the elements of the second set.

Problem 3. Prove that $3^n \geq n^3$ for all positive integers n .

Problem 4. Let $n \geq 6$ be an integer. Show that

$$\left(\frac{n}{3}\right)^n < n! < \left(\frac{n}{2}\right)^n.$$

Problem 5. Given 50 distinct positive integers strictly less than 99, prove that some two of them sum to 99.

Problem 6. In each of the unit squares of a 10×10 checkerboard, a positive integer not exceeding 10 is written. Any two numbers that appear in adjacent or diagonally adjacent squares of the board are relatively prime. Prove that some number appears at least 17 times.

Problem 7. Given $n \geq 3$ points in the plane, prove that some three of them form an angle less than or equal to $\frac{\pi}{n}$.